Biodiesel is a fuel made from vegetable oils, fats or greases. Biodiesel fuel can be used in diesel engines without altering the engine. Pure biodiesel is non-toxic, biodegradable and produces lower level of air pollutants than petroleum-based diesel fuel. The Government of India approved the National Policy on Biofuels in December 2009 and identified *Jatropha curcas* as the most suitable oilseed for biodiesel production. *Pongamia* species is also a suitable choice for production of biodiesel.

$$C_6H_{12}O_6 \xrightarrow[\text{Fermentation}]{\text{Yeast}} 2C_2H_5OH + 2CO_2\uparrow$$
(Glucose) (Ethanol)



World biofuel day is observed every year on **10th August** to create awareness about the importance of renewable bio-fuels as an alternative to conventional non-renewable fossil fuels. This day also highlights the various efforts taken by the Government in the biofuel sector.

9.2.3 Chemicals, enzymes and other bioactive molecules

Microbes are not only used for commercial and industrial production of alcohol, but also used for production of chemicals like organic acids and enzymes. Examples of organic acid producers are *Aspergillus niger* for **citric acid**, *Acetobacter aceti* for **acetic acid**, *Rhizopus oryzae* for **fumaric acid**, *Clostridium butyricum* for **butyric acid** and *Lactobacillus* for **lactic acid**.

Yeast (*Saccharomyces cerevisiae*) and bacteria are used for commercial production of enzymes. Lipases are used in detergent formulations and are used for removing oily stains from the laundry. Bottled juices are clarified by the use of **pectinase, protease** and **cellulase**. Rennet can also be used to separate milk into solid curds for cheese making. Streptokinase produced by the bacterium *Streptococcus* and genetically engineered *Streptococci* are used as "**clot buster**" for removing clots from the blood vessels of patients who have undergone myocardial infarction.

Cyclosporin A, an immunosuppressant used in organ transplantation is produced from the fungus *Trichoderma polysporum*. It is also used for its anti-inflammatory, anti-fungal and anti-parasitic properties. **Statins** produced by the yeast *Monascus purpureus* have been used to lower blood cholesterol levels. It acts by competitively inhibiting the enzyme responsible for the synthesis of cholesterol. Recombinant **human insulin** has been produced predominantly using *E. coli* and *Saccharomyces cerevisiae* for therapeutic use in human.

9.3 Microbes in sewage treatment and energy generation

Sewage is the waste generated every day in cities and towns containing human excreta. It contains large amounts of organic matter and microbes, which are pathogenic to humans and are bio-degradable pollutants. Domestic waste consists of approximately 99 percent water, suspended solids and other soluble organic and inorganic substances. Sewage should not be discharged directly into natural water bodies like rivers and streams. Before disposal, sewage should be treated in sewage treatment plants to make it less polluting.

Microbes(mass of bacteria floc) are allowed to grow in aerated water (secondary treatment). They consume major part of organic matter in the effluent and reduce the BOD in the waste water(The details on waste water treatment are discussed in chapter 13).

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